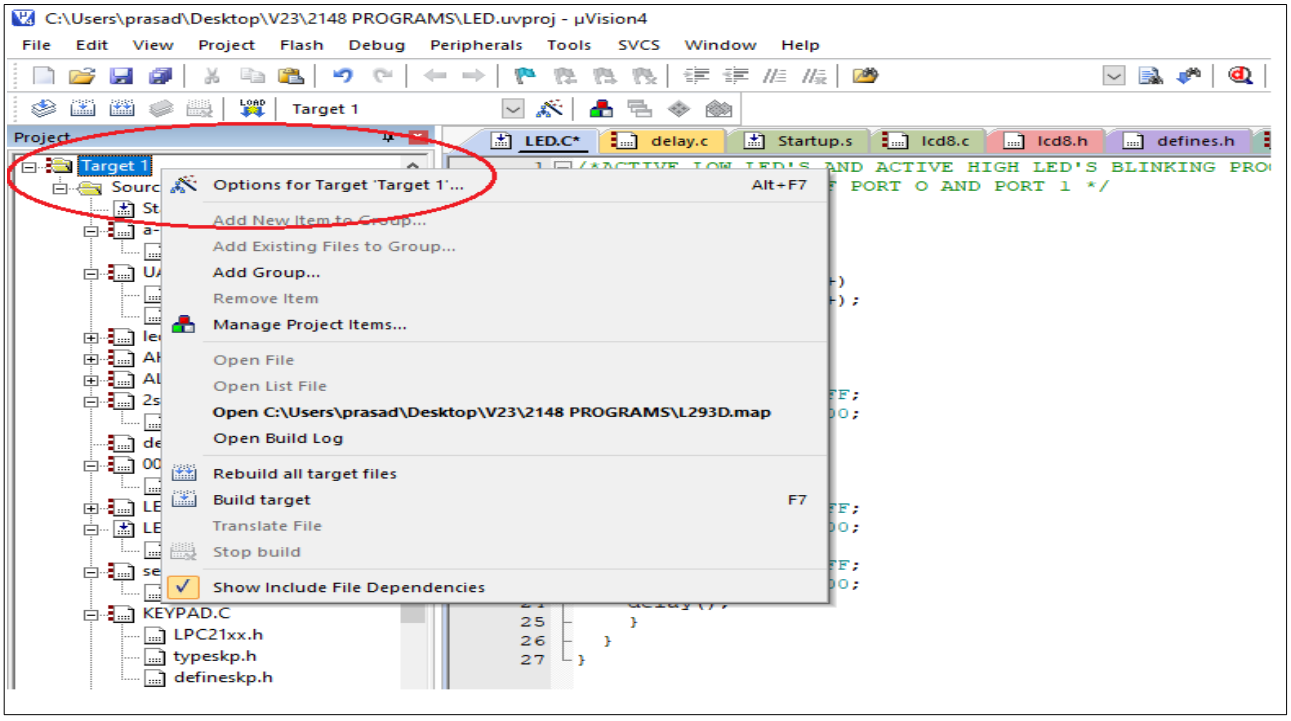
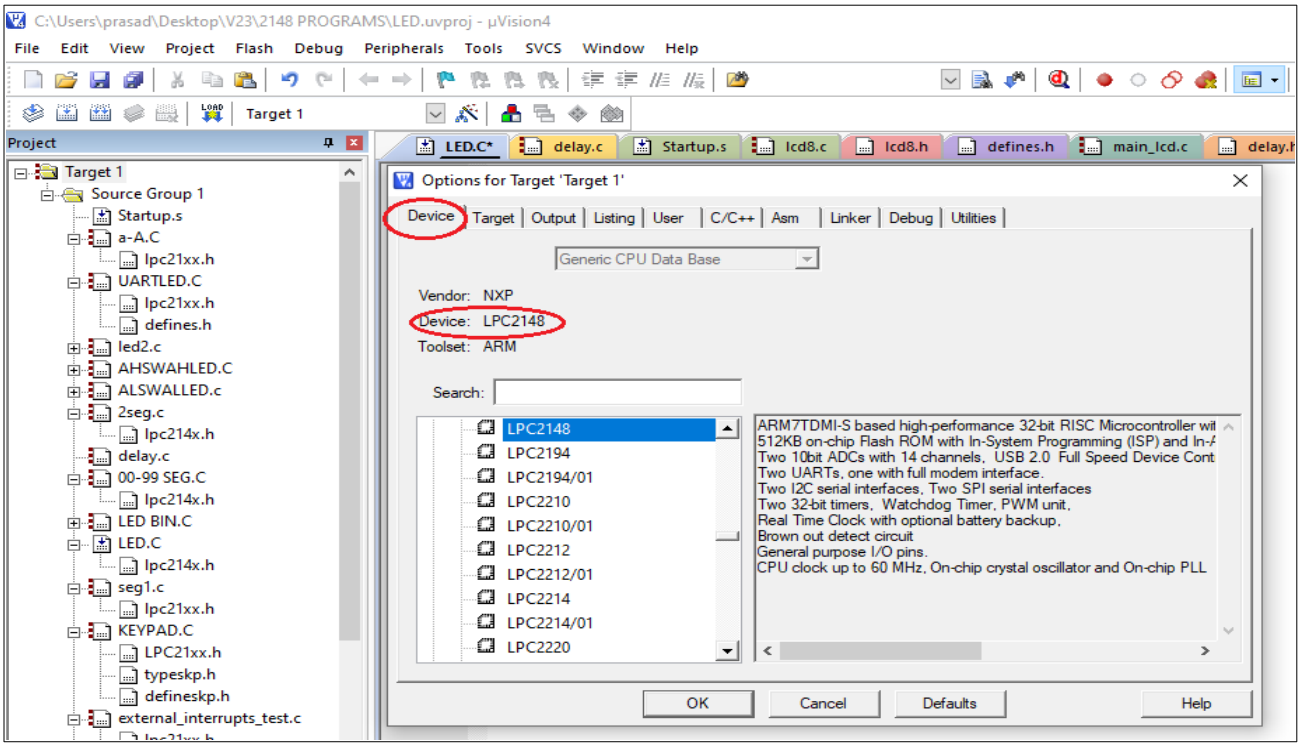
HOW TO CREATE HEX FILE:

1. RIGHT CLICK ON TARGET IN PROJECT WINDOW.

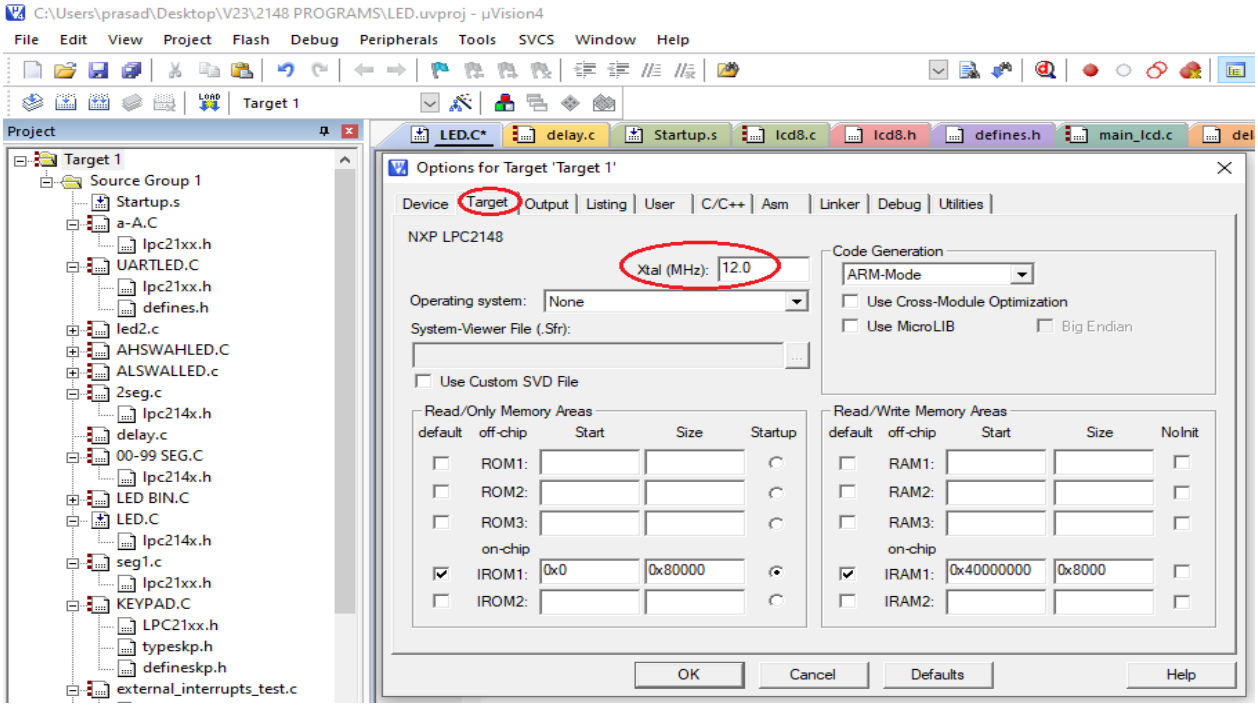
2. SELECT OPTIONS FOR TARGET



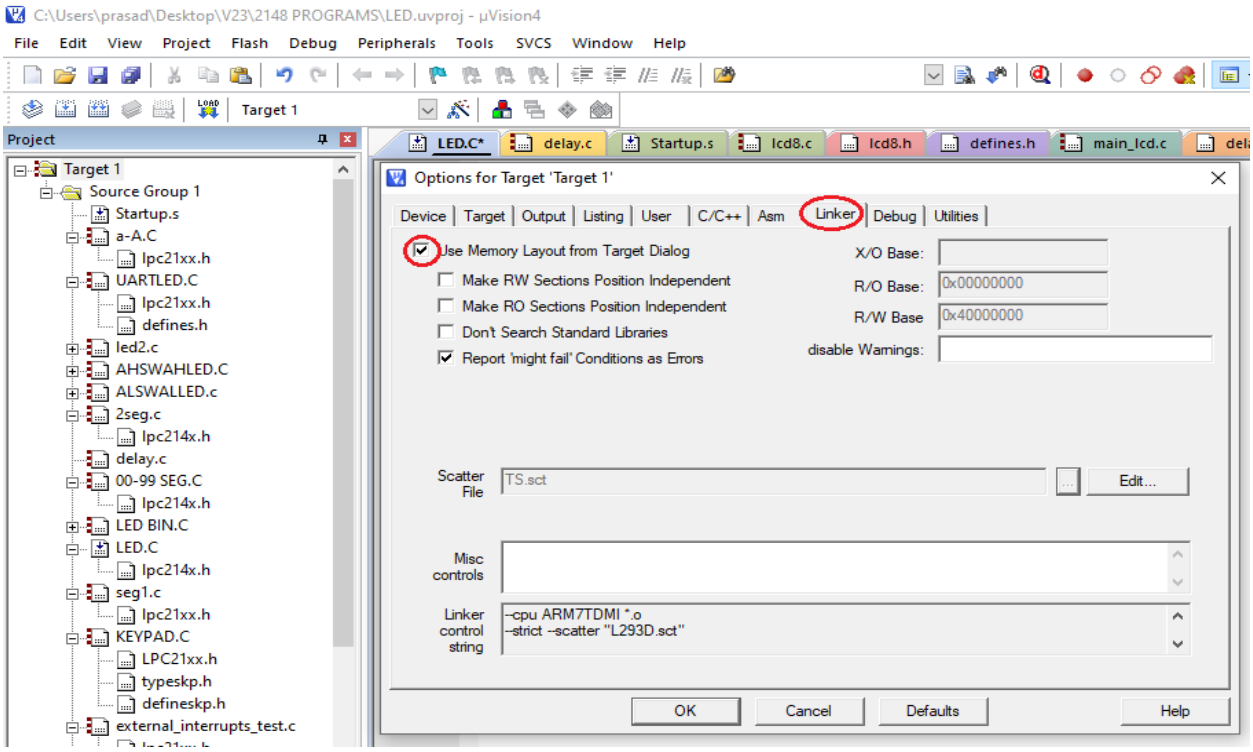
3. CLICK ON DEVICE, DEVICE SHOULD BE LPC2148



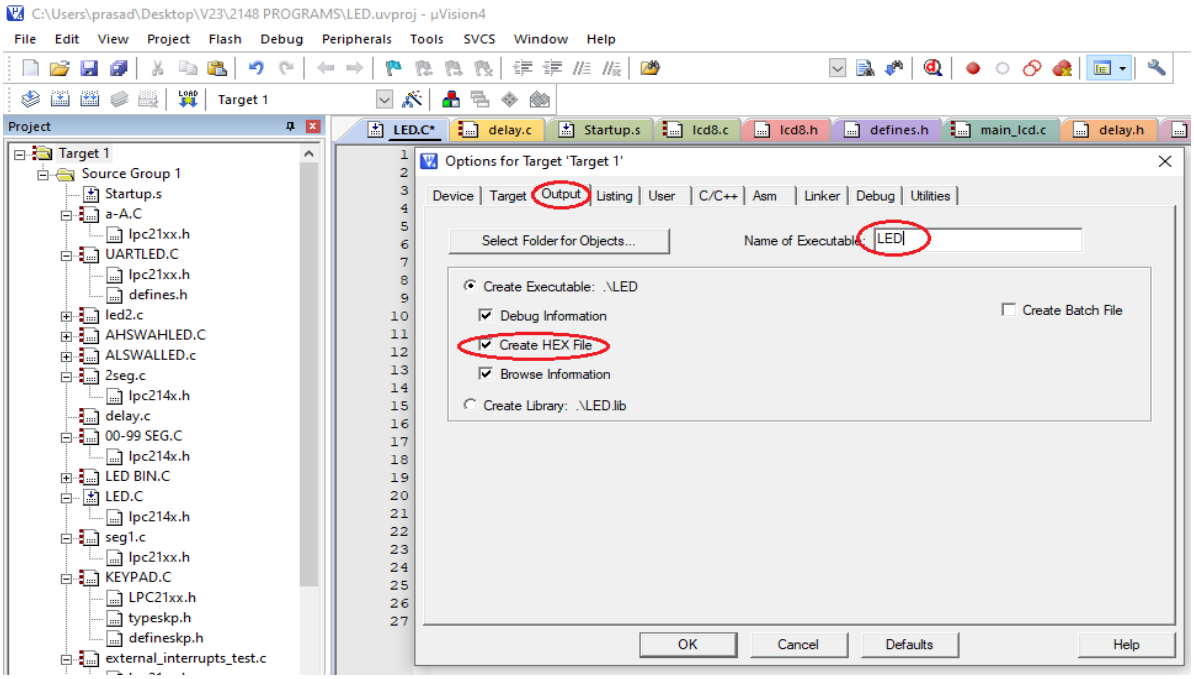
4. CLICK ON TARGET, Xtal (MHz)=12



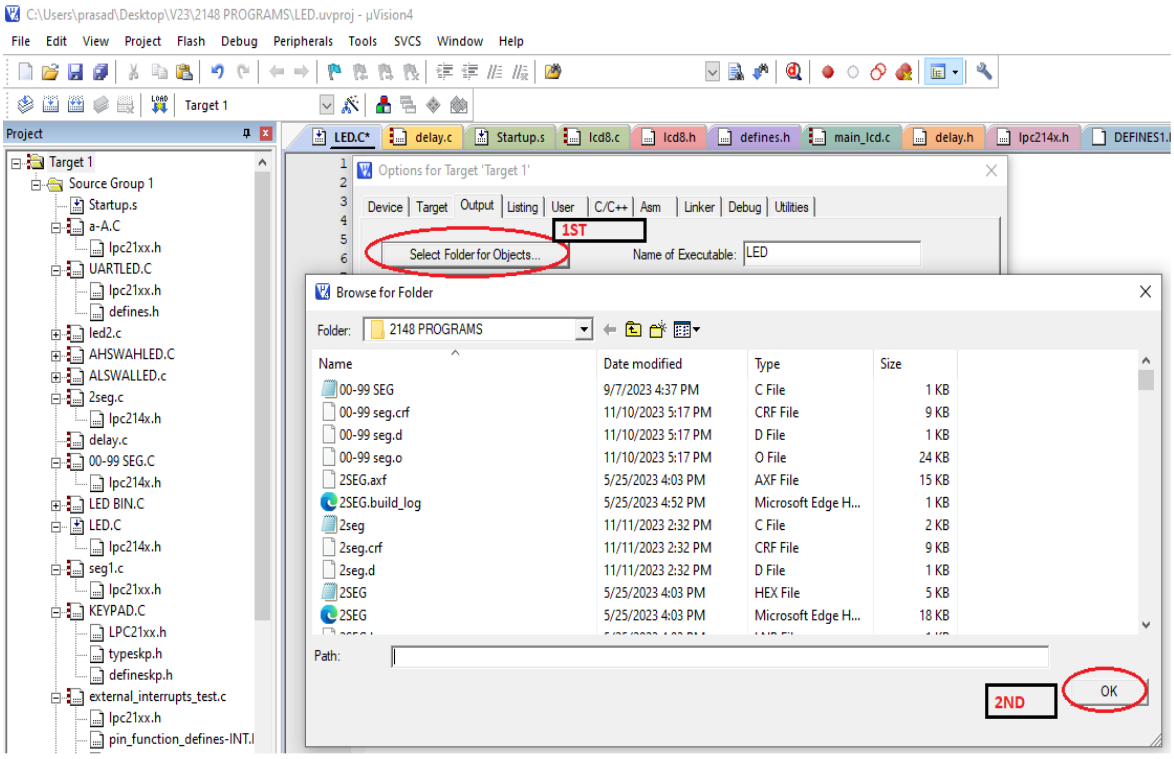
5. CLICK ON LINKER, THEN ENABLE USE MEMORY LAYOUT.



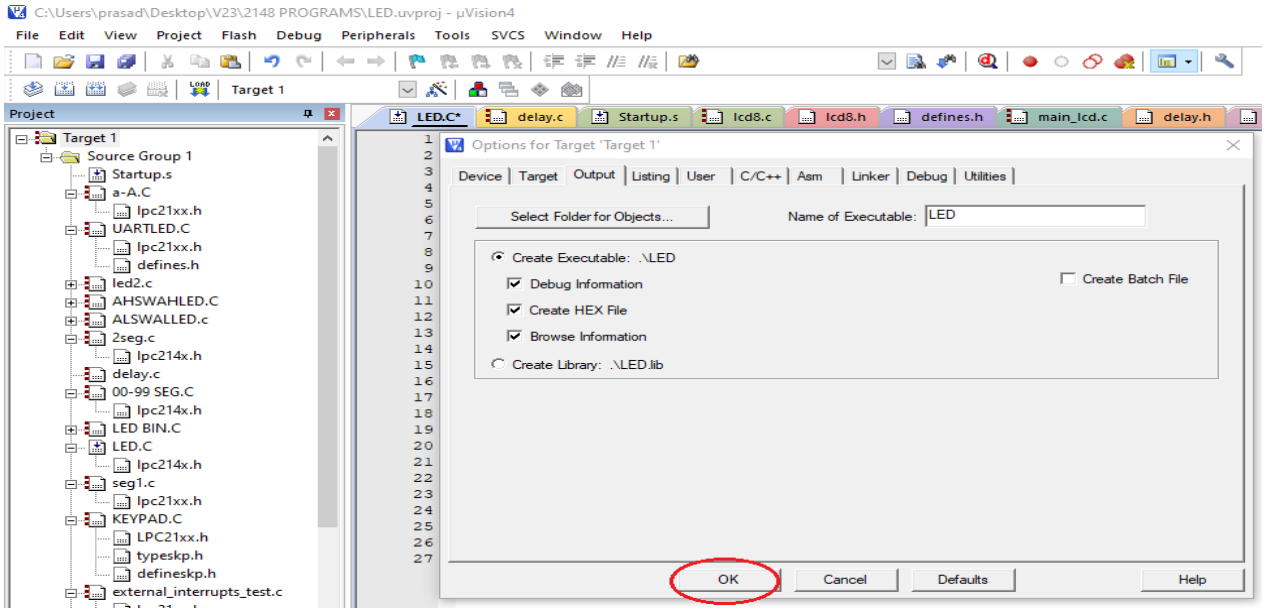
6. CLICK ON OUTPUT, THEN ENABLE CREATE HEX FILE AND GIVE HEX FILE NAME AT " NAME OF EXECUTABLE".



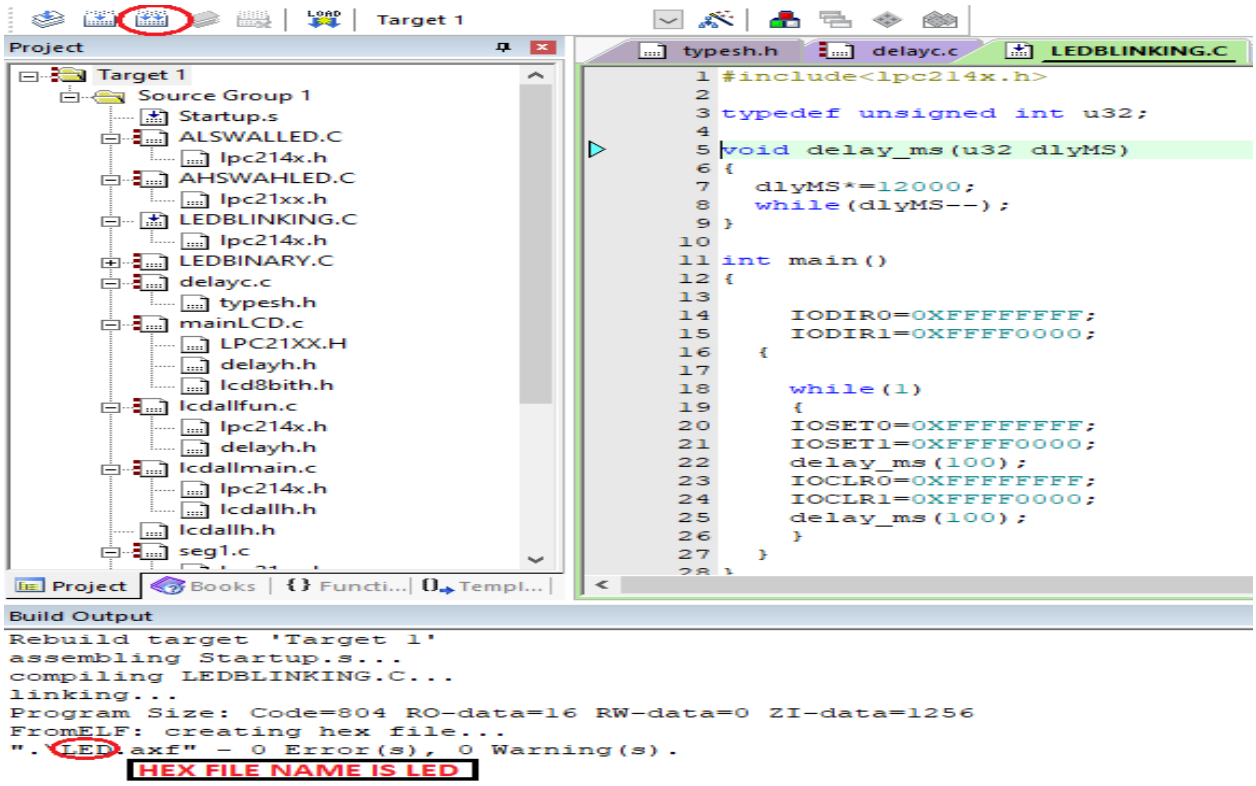
7. CLICK ON FIRST SELECT FOLDER FOR OBJECTS. USER HAVE TO SELECT PATH, PROJECT FILE AND HEX FILE SHOULD BE IN SINGLE FOLDER. FINALLY CLICK ON OK.



8. CLICK ON OK.



9. CLICK ON REBUILD. BEFORE GOING TO LOAD HEX FILE USER SHOULD REBUILD THE PROGRAM.



HOW TO USE ARM7 DEVELOPMENT BOARD

1. CONNECT THE USB MALE TO MALE CABLE FROM USB PORT OF CPU TO USB PORT OF POWER SUPPLY BLOCK ON ARM7 BOARD.

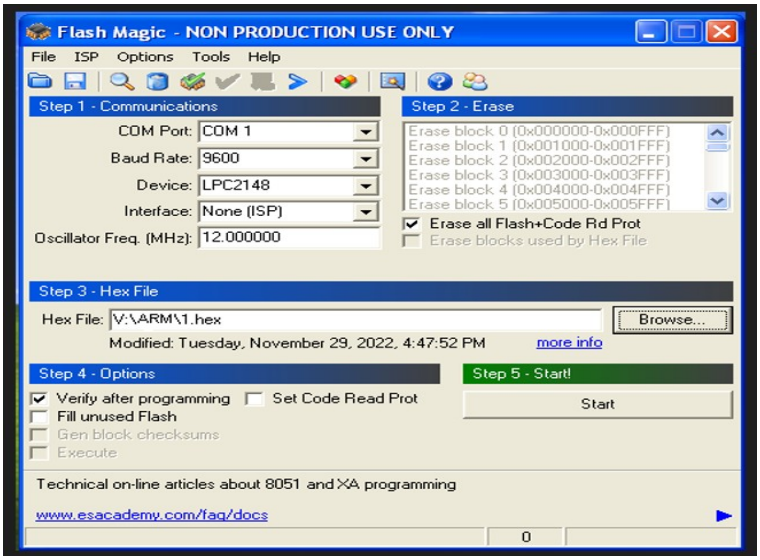
2. CONNECT UART CABLE TO UART0 OF MAX232 BLOCK FOR LOADING HEX FILE FROM FLASH MAGIC.

3. SWITCH ON THE POWER BUTTON IN POWER SUPPLY BLOCK FOR LOADING AND EXECUTION OF THE PROGRAM.

4. DON'T CONNECT ANY CONNECTORS AT P0.0 AND P0.1 WHILE LOADING HEX FILE. USER CAN USE P0.0 AND P0.1 PINS AFTER LOADING HEX FILE.

5. KEEP ISP SLIDE SWITCH IN LOAD MODE FOR LOADING THE HEX FILE INTO LPC2148, THEN PRESS AND RELEASE RESET(RST) PUSH BUTTON (OR) POWER OFF AND POWER ON IN POWER SUPPLY BLOCK.

6. USE FLASH MAGIC FOR HEX FILE LOADING. SELECT OPTIONS AS FOLLOWS.



7. USER HAVE TO BROWSE THE HEX FILE FROM PROJECT FOLDER IN STEP 3.

8. FINALLY CLICK ON START IN STEP 5 OF FLASH MAGIC.

9. KEEP ISP SLIDE SWITCH IN EXE MODE FOR EXECUTING THE PROGRAM AS PER TASK, THEN PRESS AND RELEASE RESET(RST) PUSH BUTTON (OR) POWER OFF AND POWER ON IN POWER SUPPLY BLOCK.

10. BEFORE GOING TO INTERFACE THE DEVICES TO LPC2148, ONCE READ THE PIN NAMING AND DEVICE NAMING IN DEVICES BLOCK.

11. WHILE INTERFACING DEVICES SWITCH OFF THE BOARD POWER. AFTER INTERFACING DEVICES SWITCH ON BOARD POWER

12. FOR EXECUTION, CONNECT THE CONNECTORS FROM PORT PINS TO DEVICES AS PER PROGRAM PORT PINS

LED'S TESTING WITH OUT PROGRAM:

DON'T CONNECT 1 PIN CONNECTOR FROM LED'S TO OPEN SUPPLY PINS (5V,3.3, GND) IN POWER SUPPLY BLOCK FOR LED CHECKING WITH OUT PROGRAM

CONNECT RESISTOR CABLE FROM ACTIVE HIGH LED'S TO 3.3V AND ACTIVE LOW LED'S TO GND IN POWER SUPPLY BLOCK FOR LED CHECKING WITH OUT PROGRAM.

LED'S TESTING WITH PROGRAM:

TEST ON-BOARD LED'S WITH THE HELP OF LED BLINKING PROGRAM FROM LMS.

CONNECT 4 PIN CONNECTOR FROM P0.16 - P0.19 OF ARM TO ACTIVE LOW LED'S AND CONNECT 4 PIN CONNECTOR FROM P0.20 - P0.23 OF ARM TO ACTIVE HIGH LED'S.

SWITCHES TESTING WITH OUT PROGRAM:

FOR ACTIVE HIGH SWITCHES AND ACTIVE LOW SWITCHES TESTING PROCESS INITIALLY USER SHOULD TEST ACTIVE HIGH LED'S AND ACTIVE LOW LED'S WITH RESISTOR CONNECTOR.[

CONNECT RESISTOR CABLE FROM ACTIVE HIGH LED'S TO 3.3V AND ACTIVE LOW LED' S TO GND IN POWER SUPPLY BLOCK FOR LED CHECKING WITH OUT PROGRAM.

CONNECT THE RESISTOR CABLE FROM ACTIVE LOW SWITCH TO ACTIVE LOW LED ,IF USER PRESS ACTIVE LOW SWITCH THEN ACTIVE LOW LED ON MEANS ACTIVE LOW SWITCH IS WORKING. IF ACTIVE LOW LED IS NOT GLOW MEANS ACTIVE LOW SWITCH IS NOT WORKING.

CONNECT THE RESISTOR CABLE FROM ACTIVE HIGH SWITCH TO ACTIVE HIGH LED ,IF USER PRESS ACTIVE HIGH SWITCH THEN ACTIVE HIGH LED ON MEANS ACTIVE HIGH SWITCH IS WORKING. IF ACTIVE HIGH LED IS NOT GLOW MEANS ACTIVE HIGH SWITCH IS NOT WORKING.

SWITCHES TESTING WITH PROGRAM:

LOAD THE HEX FILE ACTIVE LOW SWITCH WITH ACTIVE LOW LED AND TEST THE ACTIVE LOW SWITCHES WITH ACTIVE LOW LED'S AS PER PROGRAM PORT PINS

IF USER PRESS THE ACTIVE LOW SWITCH, ACTIVE LOW LED SHOULD ON.
IF LED'S IS NOT GLOW MEANS SWITCH IS NOT WORKING(USER SHOULD TEST ACTIVE LOW LED WITH RESISTOR WIRE).

LOAD THE HEX FILE ACTIVE HIGH SWITCH WITH ACTIVE HIGH LED AND TEST THE ACTIVE HIGH SWITCHES WITH ACTIVE HIGH LED'S AS PER PROGRAM PORT PINS.

IF USER PRESS THE ACTIVE HIGH SWITCH, ACTIVE HIGH LED SHOULD ON.
IF LED'S ARE NOT GLOW MEANS SWITCH IS NOT WORKING(USER SHOULD TEST ACTIVE HIGH LED WITH RESISTOR WIRE).

COMMON ANODE SEGMENT TESTING WITH OUT PROGRAM:

SEG1 TESTING:

CONNECT RESISTOR CONNECTOR FROM 3.3V TO SEG1 PIN OF 7 SEGMENT BLOCK

CONNECT 1 PIN CONNECTOR FROM GND TO A TO DP PINS OF 7 SEGMENTS BLOCK.

SEG2 TESTING:

CONNECT RESISTOR CONNECTOR FROM 3.3V TO SEG2 PIN OF 7 SEGMENT BLOCK

CONNECT 1 PIN CONNECTOR FROM GND TO A TO DP PINS OF 7 SEGMENTS BLOCK.

COMMON ANODE SEGMENT TESTING WITH PROGRAM:

SEG1 TESTING:

LOAD LED BLINKING HEX FILE (FROM LMS) IN TO ARM.

CONNECT 8 PIN CONNECTOR FROM P0.16 - P0.23 OF ARM TO A-DP OF 7 SEGMENT BLOCK.

CONNECT 1 PIN CONNECTOR FROM 3.3V OF MAX232 BLOCK TO SEG1 PIN OF 7 SEGMENTS BLOCK.

SEG2 TESTING:

LOAD LED BLINKING HEX FILE (FROM LMS) IN TO ARM.

CONNECT 8 PIN CONNECTOR FROM P0.16 - P0.23 OF ARM TO A-DP OF 7 SEGMENT BLOCK.

CONNECT 1 PIN CONNECTOR FROM 3.3V OF MAX232 BLOCK TO SEG2 PIN OF 7 SEGMENTS BLOCK.

LCD TESTING:

CONNECT RS AND EN PINS FROM LCD BLOCK TO PORT PINS (WHICH PINS USED IN PROGRAM BY USER)

CONNECT D0 TO D7 PINS FROM LCD BLOCK TO PORT PINS (WHICH PINS USED IN PROGRAM BY USER)

IF USER NOT ABLE TO SEE DATA ON LCD, VARY THE VARIABLE RESISTOR (YELLOW COLOR) AND CROSS CHECK THE CONNECTOR COLORS AS PER PROGRAM PORT PINS IN LCD BLOCK.